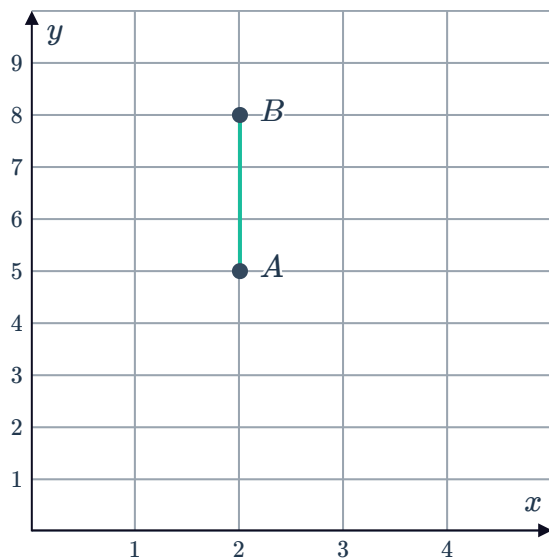


## Distance between two points

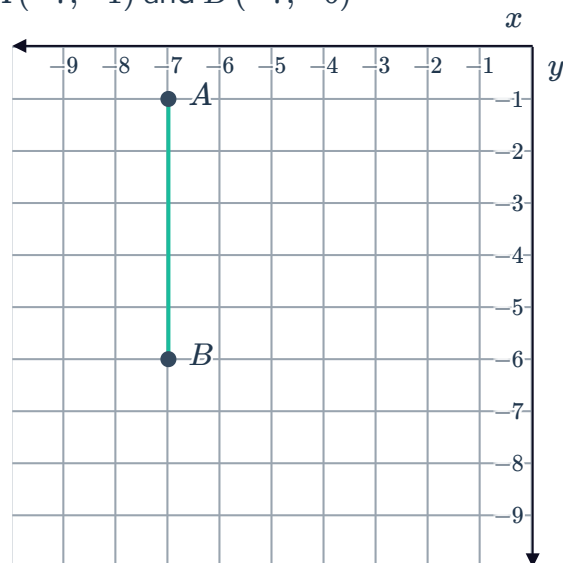


1 Find the length of each vertical interval on the number planes:

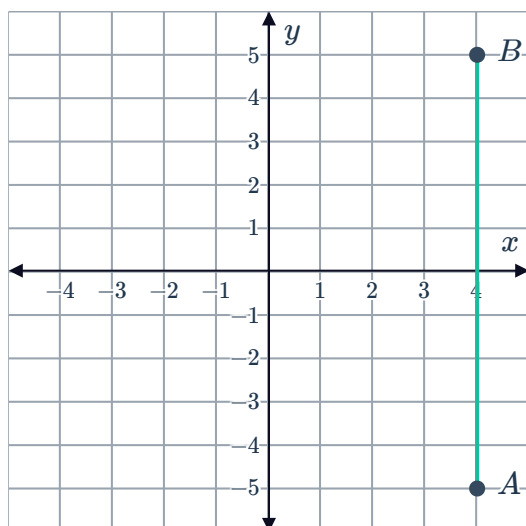
a  $A(2, 5)$  and  $B(2, 8)$



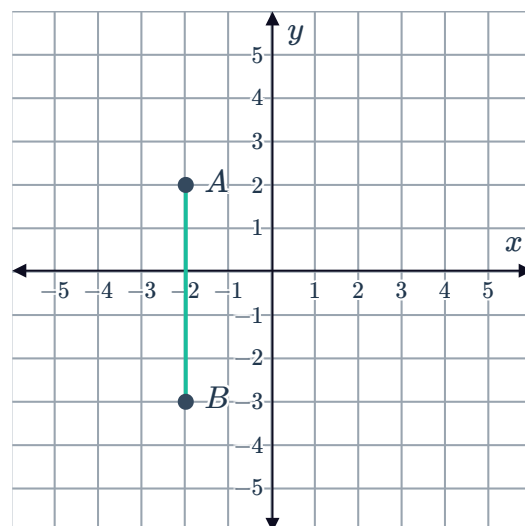
b  $A(-7, -1)$  and  $B(-7, -6)$



c  $A(4, -5)$  and  $B(4, 5)$



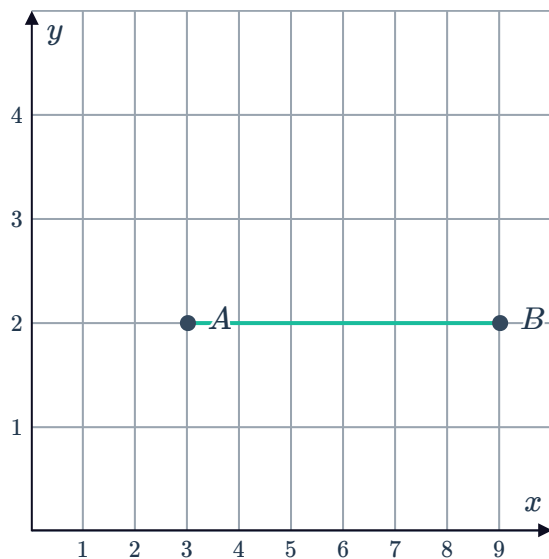
d  $A(-2, 2)$  and  $B(-2, -3)$



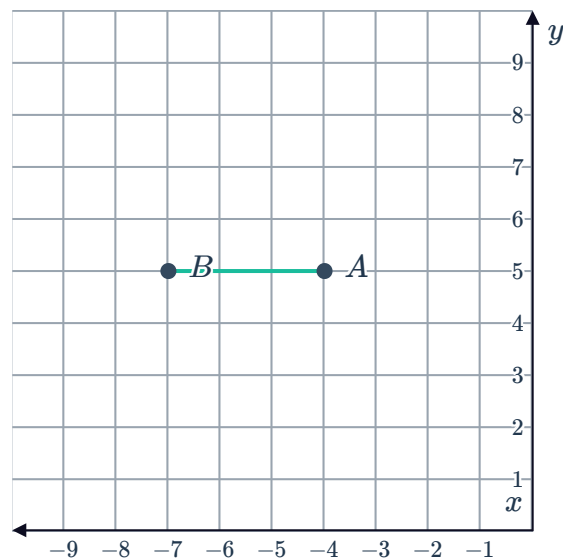
2 Find the length of each horizontal interval on the number planes:



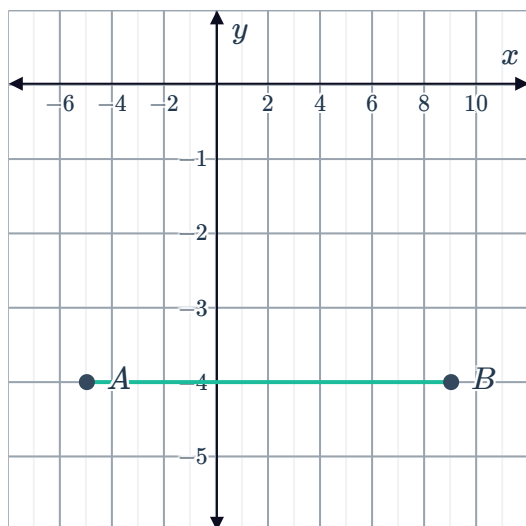
a  $A(3, 2)$  and  $B(9, 2)$



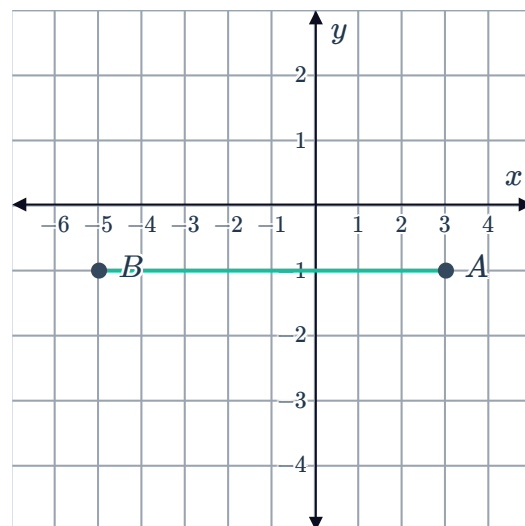
b  $A(-4, 5)$  and  $B(-7, 5)$



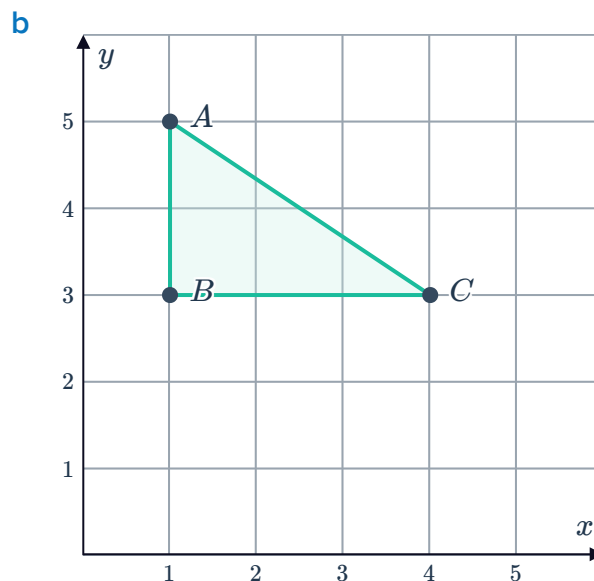
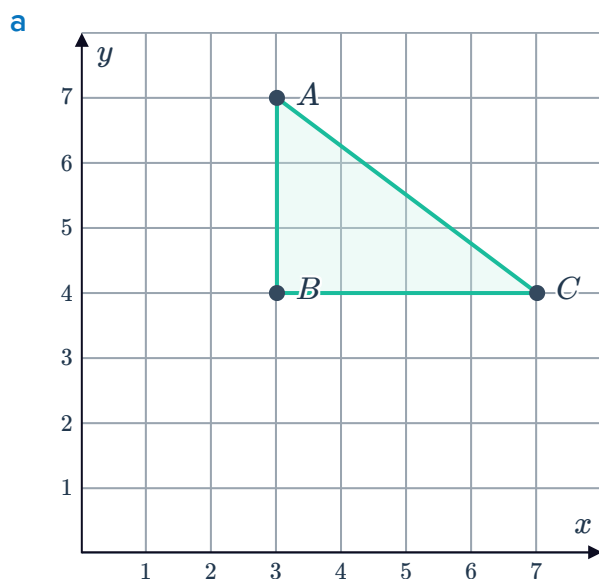
c  $A(-5, -4)$  and  $B(9, -4)$



d  $A(3, -1)$  and  $B(-5, -1)$



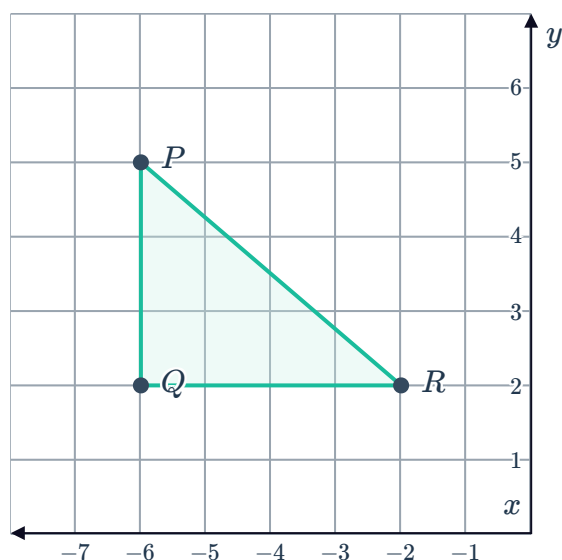
- 3 For each of the following right-angled triangles  find the length of side  $AC$ . Round your answer to two decimal places if necessary.



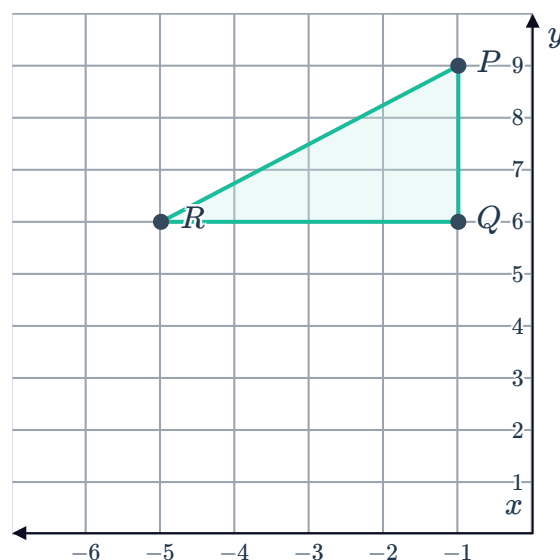
- 4 For each of the following right-angled triangles:

- Find the length of interval  $PQ$ .
- Find the length of interval  $QR$ .
- Find the length of  $PR$ , rounding to two decimal places if necessary.

- a  $P(-6, 5)$ ,  $Q(-6, 2)$  and  $R(-2, 2)$



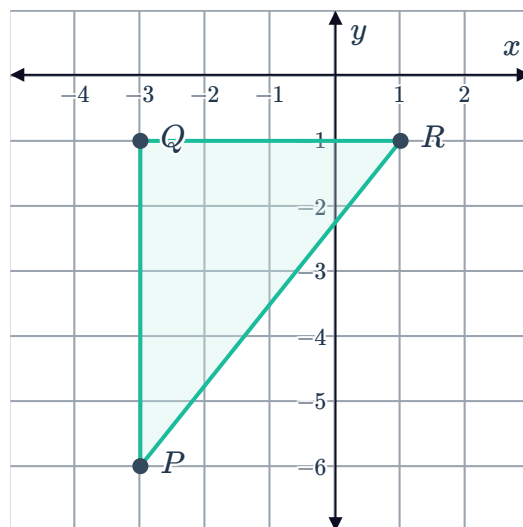
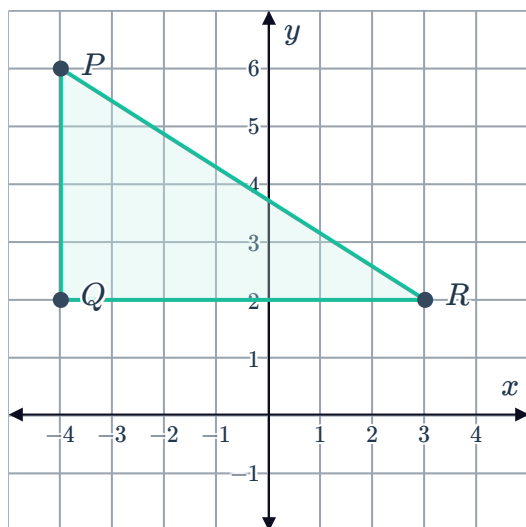
- b  $P(-1, 9)$ ,  $Q(-1, 6)$  and  $R(-5, 6)$



c  $P(-4, 6)$ ,  $Q(-4, 2)$  and  $R(3, 2)$



d  $P(-3, -6)$ ,  $Q(-3, -1)$  and  $R(1, -1)$

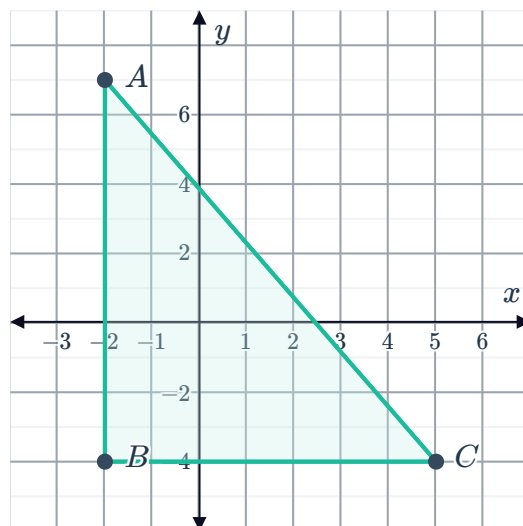
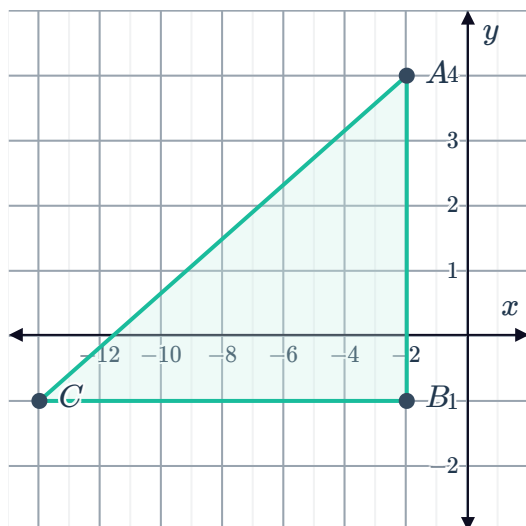


5 For each of the following right-angled triangles:

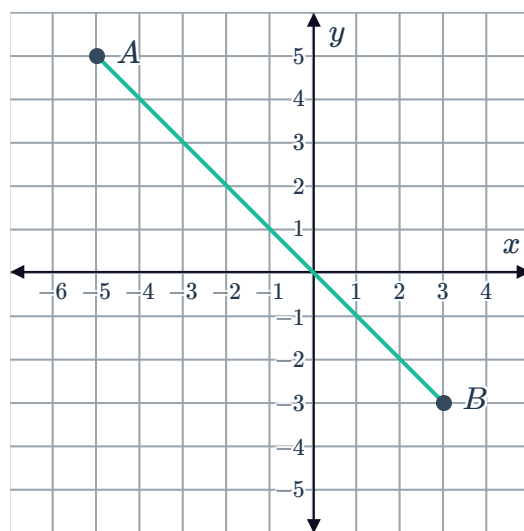
- i Find the length of interval  $AB$ .
- ii Find the length of interval  $BC$ .
- iii Find the length of  $AC$  denoted by  $c$ , rounding to two decimal places if necessary.

a  $A(-2, 4)$ ,  $B(-2, -1)$  and  $C(-14, -1)$

b  $A(-2, 7)$ ,  $B(-2, -4)$  and  $C(5, -4)$



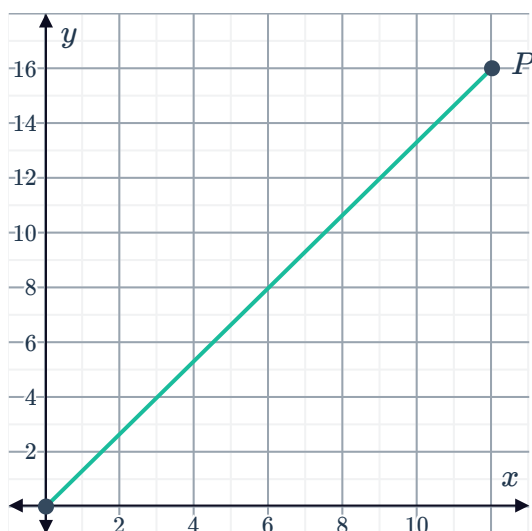
- 6 Find the length of  $AB$  shown on the graph. Round your answer to two decimal places.



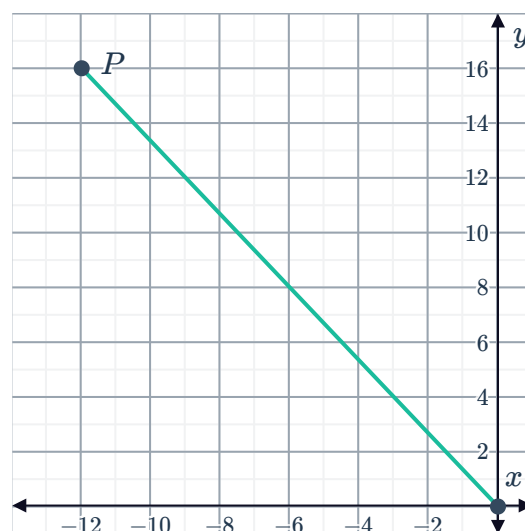
### Distance formula (Extended)

- 7 Find the distance of the given point  $P$  from the origin.

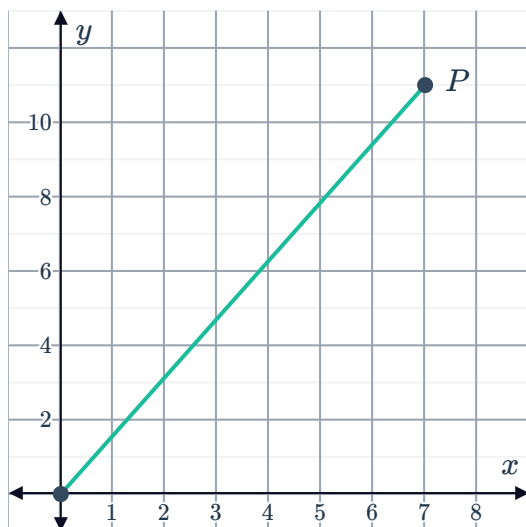
a  $P(12, 16)$



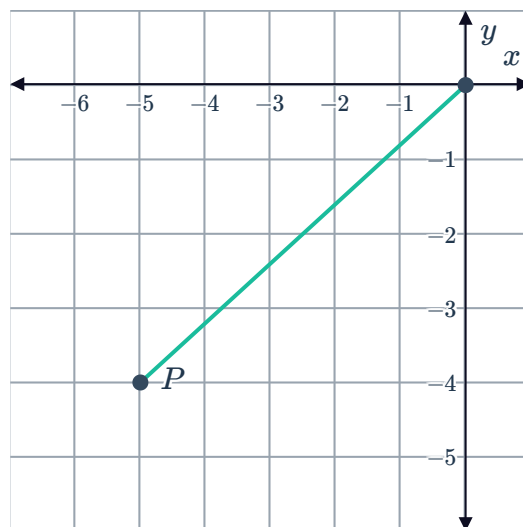
b  $P(-12, 16)$



c  $P(7, 11)$

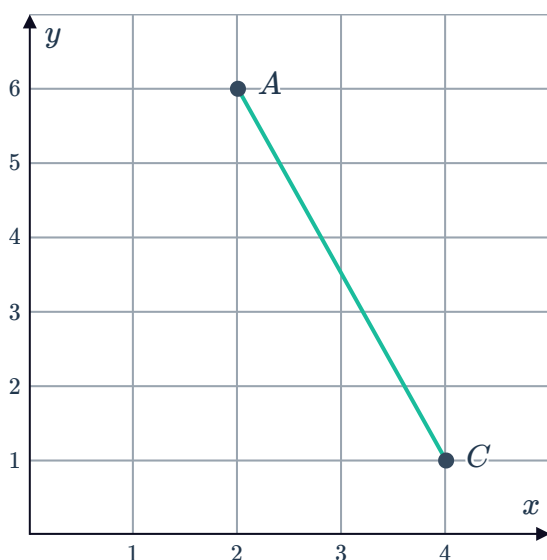


d  $P(-5, -4)$

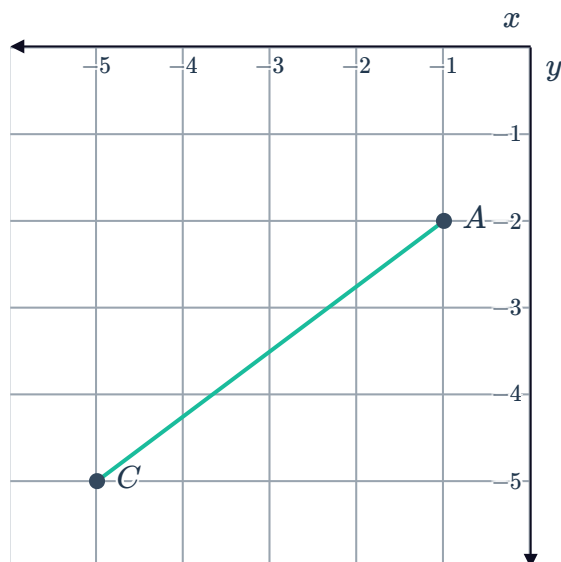


8 For each of the following graphs, find the length of the interval  $AC$ . Round your answer to two decimal places if necessary:

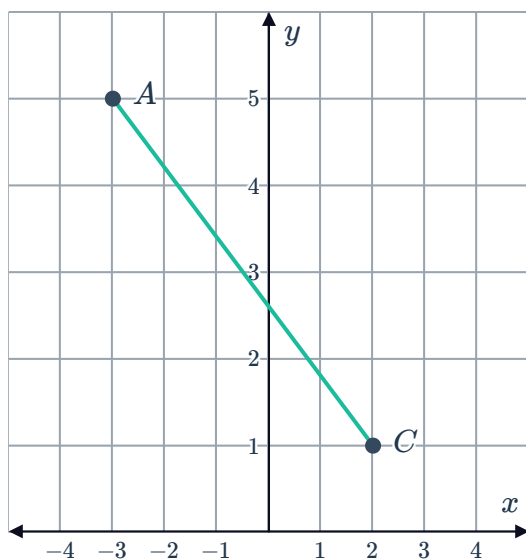
a



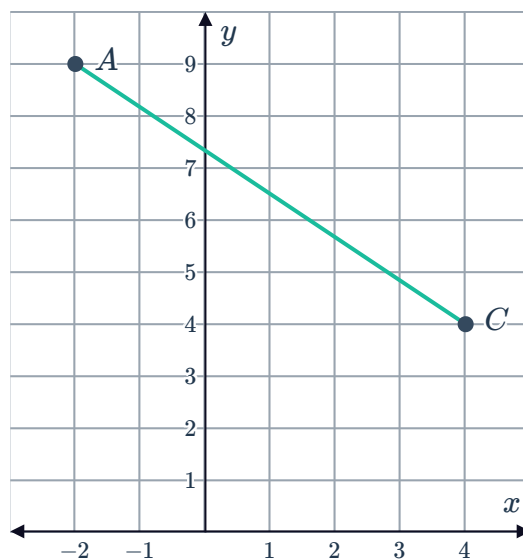
b



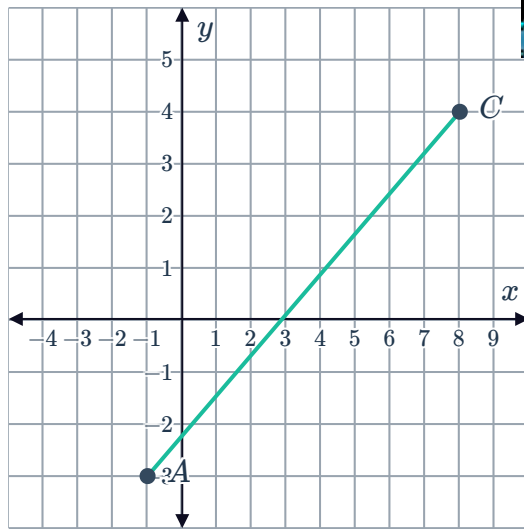
c



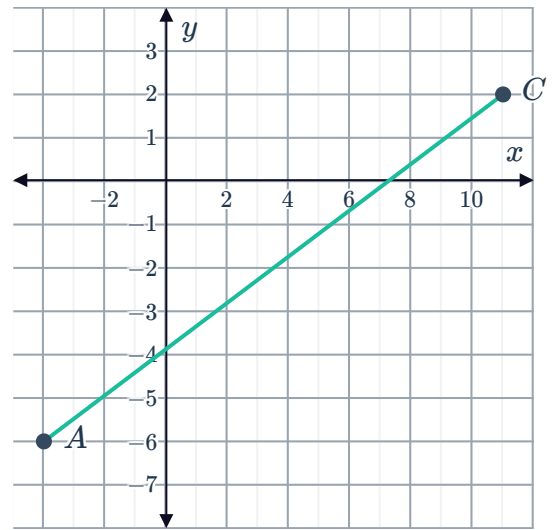
d



e



f



9 Find the distance between Point  $A$  and Point  $B$ . Write your answer in surd form if necessary.

a  $A(1, 4)$  and  $B(7, 12)$

b  $A(4, 2)$  and  $B(-8, -7)$

c  $A(-1, 9)$  and  $B(-4, 1)$

d  $A\left(-1, -\frac{3}{5}\right)$  and  $B\left(4, \frac{12}{5}\right)$

10 Consider the points  $M(-9, -1)$  and  $N(1, 5)$ .

a Find the exact distance from  $M$  to the origin.

b Find the exact distance from  $N$  to the origin.

c Which point is closer to the origin?

11 Given  $P(4, 3)$ ,  $M(-3, -4)$  and  $N(-7, 1)$ .

a Find the distance from  $P$  to  $M$  to two decimal places.

b Find the distance from  $P$  to  $N$  to two decimal places.

c Which point is further from  $P$ ?

12 Consider the Points  $A(12, 3)$  and  $B(14, 0)$ .

a Find the length of  $AB$  to two decimal places.

b If  $M$  is the midpoint of  $AB$ , find the length of  $AM$  to two decimal places.

13  $AM$  is a vertical interval 3 units long. If  $A$  is the point  $(-2, 6)$ , find two possible coordinates of  $M$ .

14 Consider the points  $A(-2, -6)$ ,  $B(4, -2)$  and  $C(1, -4)$ . Find:



- a The exact distance  $AC$ .
- b The exact distance  $BC$ .
- c What do you notice about the distance between the points?

15  $K$  is the midpoint of  $A(3, 1)$  and  $C(15, -7)$ . Find the distance from  $A$  to  $K$  correct to one decimal place.

---

## Geometric applications (Extended)

16 A triangle is formed by three points:  $A(4, -3)$ ,  $B(1, 0)$  and  $C(7, 0)$ . Find the following:

- a The distance  $BC$ .
- b The exact distance  $AB$ .
- c The exact distance  $AC$ .
- d Is this triangle equilateral, isosceles or scalene?

17  $ABCD$  is a rhombus whose vertices are  $A(1, 2)$ ,  $B(3, 10)$ ,  $C(11, 12)$  and  $D(9, 4)$ . Find:

- a The exact length of diagonals:
  - i  $AC$
  - ii  $BD$
- b The exact area of the rhombus.

18 A circle with centre at point  $C(3, 4)$  has point  $A(-12, 12)$  lying on its circumference. Find:

- a The radius of the circle.
- b The exact circumference of the circle.
- c The distance between point  $(20, 4)$  and the centre.
- d Does the circle also pass through the point  $(20, 4)$ ? Explain your answer.

19 A triangle has vertices at  $A(1, -1)$ ,  $B(-3, -4)$  and  $C(5, -4)$ .

- a Find the length of the following sides:
  - i  $AB$
  - ii  $AC$
  - iii  $BC$
- b Is this triangle equilateral, isosceles or scalene?



**20** Consider the triangle whose vertices are   $A(-2)$ ,  $B(2, -8)$  and  $C(3, 5)$ .

- a** How can we show that the triangle has a  $90^\circ$  and two  $45^\circ$  angles?
- b** Find the length of the following sides in exact form:
  - i**  $AB$
  - ii**  $AC$
  - iii**  $BC$
- c** Does the triangle have a  $90^\circ$  and two  $45^\circ$  angles?

**21**  $\triangle PQR$  has vertices  $P(2, -6)$ ,  $Q(-9, -17)$  and  $R(-5, 1)$ .

- a** Find the length of the following sides in exact form:
  - i**  $PQ$
  - ii**  $QR$
  - iii**  $PR$
- b** Is the triangle right-angled?

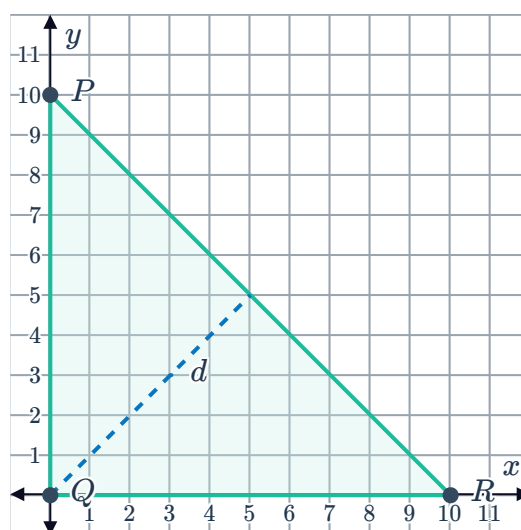
**22** The points  $M(-3, -5)$  and  $R(4, -12)$  are the end points of an interval.  $N(-1, -7)$  is a point on this interval.

- a** Find the exact distance between the following points:
  - i**  $M$  and  $N$
  - ii**  $N$  and  $R$
- b** In what ratio does  $N$  divide  $MR$ ?

**23** Find the perimeter of the parallelogram whose vertices are  $A(3, 2)$ ,  $B(-2, 7)$ ,  $C(2, 9)$  and  $D(7, 4)$ . Round your answer correct to one decimal place.

**24** The isosceles  $\triangle PQR$  is shown on the number plane:

- a** Find the area of the triangle.
- b** Find the exact length of  $PR$ .
- c** Find the exact value of  $d$ , the perpendicular distance from  $Q$  to the side  $PR$ .



**25** Consider the points  $P(10x, -6x)$  and  $Q(4x, -2x)$ , where  $x > 0$ .

- a** Find the distance between  $P$  and  $Q$  in terms of  $x$ .
- b** Find the coordinates of the midpoint of the segment  $PQ$  in terms of  $x$ .

**26**  $P(-4, -4)$ ,  $Q(5, -7)$ ,  $R(9, -11)$  and  $S(0, -8)$  are the vertices of a quadrilateral.

- a** Find the exact length of side  $PQ$ .
- b** Find the exact length of side  $QR$ .
- c** Find the exact length of side  $RS$ .
- d** Find the exact length of side  $SP$ .
- e** Find the exact length of the diagonal  $PR$ .
- f** Find the exact length of the diagonal  $QS$ .
- g** Classify the quadrilateral.